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Survival and Growth of Isolated Polyps of *Galaxea fascicularis* (Linnaeus 1767) on Six Kinds of Culture Substrates: Implications for Mariculture, Aquarium Culture, and Conservation

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Abstract

Coral aquaculture is viewed as a sustainable method of providing colonies for use in the aquarium trade, pharmacological studies, and coral reef rehabilitation. The production of large, healthy specimens depends on providing conditions that mimic or surpass natural reef environments. Materials with a high degree of variability in physical and chemical properties are utilized as substrates in coral aquaculture, but until fragment–substrate interactions are isolated and independently studied, the possible inhibitory and facilitative effects of each material will remain unclear. To understand potential key roles that substrates can play in coral culture, a comparison was made about the role of various materials on fragment survival and growth. The experiment examined the interactions between isolates of a common scleractinian coral, *Galaxea fascicularis*, and six commonly used substrates (coral rock, terracotta, polyvinyl chloride, steel, rubber tire, and concrete) and how these affected survival and growth. No differences were observed in the survival, colony size, and polyp production of the resulting microcolonies. The results show that, for *G. fascicularis*, growth is not affected by substrate type. The results demonstrate the use of a wide variety of materials for the successful aquaculture of scleractinian corals.

KEYWORDS

coral culture, coral mariculture, culture substrates, *Galaxea fascicularis*

The sustainable, large-scale culture of coral fragments is viewed as one of the promising developments in ensuring the continued survival of coral reefs (Bowden-Kerby 2001; Shafir et al. 2006; Bongiorni et al. 2011; Rinkevich 2014). Coral fragmentation and propagation can be used in seeding artificial reefs or damaged coral reefs without continued dependence on wild donor colonies, while microcolonies supplied into the aquarium industry lessen and even remove the pressure on coral reefs by reducing the need to collect colonies from the wild.

Although *in situ* and *ex situ* propagation of coral fragments is already occurring, methods are far from established (Rocha et al. 2015). Few studies have directly focused on the biology of coral culture and how corals react and interact with biotic and abiotic cues from a culture standpoint. Studies have focused on the effects of light (Al-Moghrabi et al. 1995; Reynaud-Vaganay et al. 2001; Grottoli 2002; Todd et al. 2004a; Tentori and Allemand 2006; Schlacher et al. 2007), nutrition (Al-Moghrabi et al. 1995; Grottoli 2002; Grover et al. 2002; Ferrier-Pagès et al. 2003; Borell et al. 2008; van Os et al. 2012; Rocha et al. 2013a, 2013b,

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2013c), seawater composition (Atkinson et al. 1995; Grover et al. 2002), flow (Todd et al. 2004b; Wijgerde et al. 2012b), temperature (Borell et al. 2008), and sedimentation (Todd et al. 2004b), but little investigation has been conducted concerning the direct and indirect effects of substrate composition on the survival and growth of coral fragment cultured on them.

Considering that coral fragments are initially cultured on prepared substrates and are eventually permanently attached and affixed onto substrates either in aquariaor ocean nurseries or onto artificial reef structures, detailed observations on substrate–fragment interactions are important in determining how these affect coral growth. For practicality, necessity, and economy, a wide variety of materials have been utilized as substrates for corals in rehabilitation, research, and aquaculture. These have come to compose a broad group that includes reef- and calcium-based materials such as coral rock; semifossilized corals; mollusk shells and marble; various kinds of plastics, clay, and clay/ceramic/terra-cotta derivatives; concrete and concrete-aggregate mixes; metals; glass; polystyrene; and rubber tires. Although numerous studies have been performed using these substrates in studies in coral biology, many focused on other aspects of coral biology, such as larval settlement and recruitment (Heyward et al. 2002), or utilized the substrates only as mounting media (Yap et al. 1998; Muko et al. 2000; Bowden-Kerby 2001; Raymundo 2001; Soong and Chen 2003; Todd et al. 2004a; Bielmyer et al. 2010; Guest et al. 2011; Schutter et al. 2011). Studies that did focus on studying and comparing substrate–fragment interactions either utilized only a few materials at one time (Yap and Gomez 1984, 1985; Schlacher et al. 2007) or were performed in the field, in natural, uncontrolled conditions that did not isolate the direct effects of the materials on fragment survival and growth (Yap and Gomez 1984, 1985; Yap 2004). Considering the wide variety of available materials, the lack of adequate knowledge about the effect of substrate type on the survival and growth of coral fragments impairs the success of coral propagation projects. Most fragments eventually attach to available substrates (Bruckner and Bruckner 2001; Okubo et al. 2005; Williams and Miller 2009; Guest et al. 2011), but the physical and chemical properties of utilized materials may impede or facilitate the survival and growth of cultured organisms (Spieler et al. 2001). Concrete releases calcium and increases alkalinity in seawater during the curing process (Berry et al. 1977; Garrabrants et al. 2003; Gervais et al. 2004; Sani et al. 2005) and should theoretically be a suitable material for coral fragments. On the other hand, rubber tires are known to leach numerous substances (Wik 2007), including trace amounts of oil and polycyclic aromatic polycarbons (Stephensen et al. 2003) and metals such as zinc, cadmium, copper, and lead (Nelson et al. 1994; Gualtieri et al. 2005) after immersion. Steel and other metals oxidize in contact with seawater (Refait et al. 2003; Melchers and Jeffrey 2005; Zou et al. 2011) and may release byproducts that can affect water chemistry. Corals were adversely affected by these factors in long-term, high concentration experiments (Mitchelmore et al. 2007; Sabdono 2009), but the extent of these effects on small coral fragments in amounts relative to captive culture methods remains undetermined and unquantified.

The aim of the study was to perform an experiment that utilized various types of commonly used substrates of a similar size and shape, using a single animal model, performed at the same time in controlled laboratory conditions. It attempted to determine the survival and growth performance of single polyp isolates of the scleractinian coral, *Galaxea fascicularis*, grown on six different kinds of commonly used culture substrates, with the main hypothesis being that the high variability in physical and chemical properties of the various materials and their resulting reactions with seawater would differentially affect fragment survival and subsequent microcolony growth.

Materials and Methods

Selection and Preparation of the Test Organism

The scleractinian coral, *G. fascicularis*, was selected as the test species because it is common and hardy in captivity and exhibits fast growth

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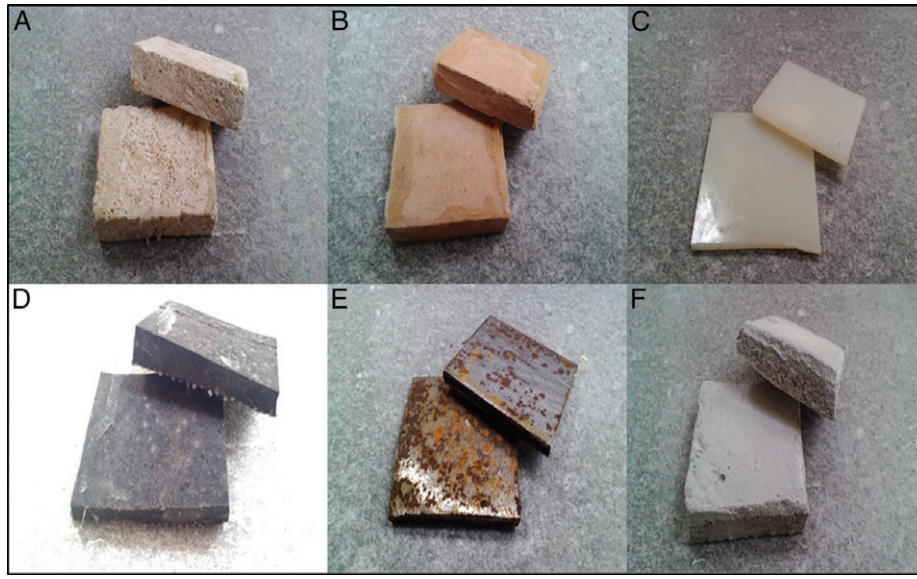

FIGURE 1. *The six different kinds of prepared substrata material used in the study. (A) mactan stone, (B) clay, (C) polyvinyl chloride, (D) rubber tire, (E) steel, and (F) concrete.*

via the production of a basal disk. It is also very amenable for the creation of single-polyp isolates at it possesses uniformly sized polyps that are easily removed from the colony's spongy coenosteum. A total of 10 colonies were collected from a nearby reef in Dumaguete, Central Philippines. Whole colonies measuring 20–25 cm were collected from flat, open reef areas at a depth of 1–2 m to closely mimic the light conditions present in the experimental setup. These were acclimated for 24 h in raceways and subsequently fragmented into individual polyps. A total of six polyps were collected from each colony, which were distributed among the six different experimental substrates. Only polyps from the central region of the colony were utilized to ensure the absence of new lateral polyps or secondary growth points. Undamaged, similarly shaped and sized polyps were used to ensure consistency and uniformity of initial samples at the start of the experiment.

Selection and Preparation of Experimental Substrates

Six substrates were selected based on a survey of commonly used materials for coral culture and artificial reef construction. These included mactan stone (mined, semifossilized coral rock), which served as the positive control, and five experimental materials that included clay (nonglazed terracotta), polyvinyl chloride (PVC), rubber tires, steel, and concrete. Tiles measuring $5 \times 5 \times 1.25$ cm were created for each of the substrates. Tiles were fashioned from larger sections of parent material (mactan stone, PVC, steel, and rubber tire) or manually prepared using established methods (clay and concrete) (Fig. 1A–F). Additional precautions and treatments were performed for some materials: nongalvanized steel was used as galvanized steel leaches zinc (Belghazi et al. 2002), which may have affected the outcome of the experiment. Concrete was allowed to cure for 30 d to prevent excessive leaching and exothermic reactions that occur during the curing process (Paulini 1990; De Schutter 2002).

Prior to use in the experiment, all tiles underwent five cycles of freshwater soaking, washing, and air drying to remove surface debris and production/fabrication byproducts.

Experimental Setup

A total of 10 polyp–substrate replicates were created for each of the six substrate types for

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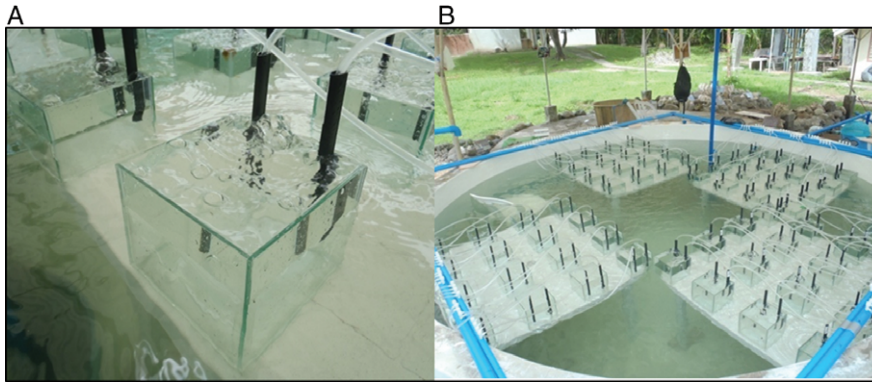

FIGURE 2. *The experimental setup. (A) Replicate tank and (B) replicate tanks immersed in water bath with air and water supply.*

a total of 60 experimental replicates. Polyps were affixed vertically by their bases onto the tiles by means of a cyanoacrylate adhesive. Tile–polyp pairs were individually housed in (15 × 15 × 15 cm) flow-through (nonrecirculating) silicone-sealed glass aquaria (Fig. 2A). Because of preset laboratory air and water flow schedules, tanks were provided with flowing seawater (ca. 300 mL/min) and aeration for 18 h and 12 h/d, respectively, with a 6-h overlap in between. Seawater and air were supplied to each tank by means of 3-mm-diameter rubber tubes tapped into water and air supply lines. Seawater with a salinity of 34–35 ppt was piped in from a nearby reef, filtered through a 25 µm mesh, and stored in a sedimentation tank prior to distribution, while air was provided by a compressor. Aquaria were randomly distributed among four plastic platforms and partially submerged in a large outdoor concrete tank that served as a water bath (Fig. 2B), insulating the tanks from wide thermal fluctuations (26.5–28.5 C) throughout the day. The setup was illuminated by natural sunlight with mean intensities of 60.10/µmol^2/sec (0900 h), 74.71/µmol^2/sec (1200 h), and 23.47/µmol^2/sec (0300 h) through clear polycarbonate roofing.

Monitoring

Each week, subsequent to mounting, all primary polyps and eventual microcolonies were examined for survival and growth. Survival was defined as the number of organisms alive during each weekly monitoring, as well as overall survival at the end of the experiment period, and was determined by observing the presence of living tissue, regardless of the presence or progression (growth) or tissue loss (recession). A polyp or microcolony was considered dead only when there was complete absence of living tissue. Growth was defined as both the increase in surface area and the production of secondary polyps of the microcolony. Increase in surface area was defined as the expansion of new soft and skeletal tissue and was thus limited to measurements of the basal disk of each primary polyp using a laboratory-grade caliper. Surface area of basal disks was determined using the formula:

$$\text{Area} = \pi \frac{\text{disk long diameter} + \text{disk short diameter}}{2}$$

Polyp production was defined as the production of secondary polyps by the original isolated polyp and eventual microcolony and was characterized as any outgrowth possessing a mouth and ring of tentacles, regardless of whether this was solely composed of soft tissue or had its own skeleton, and were recorded by manually counting discernible new growth points.

Treatment results were ranked weekly from the highest to lowest values (1 being the highest) for both mean surface area and mean polyp production (performance rank). These ranks were later averaged at the end of the study period to obtain

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the average performance ranks for each of the experimental treatments.

Statistical Analysis

Assumptions of normality and homogeneity were determined using the Shapiro–Wilk test and chi-square test, respectively. Total surface area, weekly surface area, and weekly growth rates between the six treatments were compared and analyzed using one-way ANOVA, while comparisons of total polyp counts, weekly polyp counts, and weekly polyp production rates were subjected to Kruskal–Wallis (KW) test, an analysis for nonparametric data. Appropriate post hoc tests were used to determine differences in cases where the null hypothesis was rejected. Statistical analyses were performed using version 3.16 of the Paleontological Statistics (PAST) software (Hammer et al. 2001).

Results

The observation period of the experiment lasted 10 wk, at which time the largest and fastest growing microcolonies had reached the periphery of the experimental tiles. Further growth would have been directed on a different surface and orientation, affecting measurement accuracy and quality of data. Because of tissue swelling of the newly attached polyps, basal disk measurement and polyp counts were only possible starting the third week of observation.

Survival

Survival was 100% for all polyps in all replicates of the six experimental treatments in the course of the 10-wk experimental period. Polyps and eventual microcolonies exhibited maximum soft-tissue extension and full normal coloration/pigmentation throughout the same period, with no indication of disease. No observable stress reactions, such as sweeper tentacles formation, bleaching, tissue retraction, and extracoelenteric filaments, were observed.

Colony Size

At the end of the 10-wk study period, all polyps had produced microcolonies growing on all experimental substrates (Fig. 3). Polyps grown on PVC plates exhibited the largest average surface area (7.63 cm^2; SE = 0.64, $n = 10$), followed by polyps grown on concrete (7.35 cm^2; SE = 1.02, $n = 10$), clay (7.34 cm^2; SE = 1.27, $n = 10$), and mactan stone (7.00 cm^2; SE = 0.97, $n = 10$). Polyps grown on rubber (6.20 cm^2; SE = 1.15, $n = 10$) and steel (6.20 cm^2; SE = 11.24, $n = 10$) exhibited the smallest surface area (Fig. 4A). These values, however, were not significantly different from each other (ANOVA, $P = 0.88$). Weekly comparisons of total basal tissue area growth showed fluctuations in the overall performance of polyps grown in the six experimental substrates. The largest tissue surface area shifted between polyps grown on concrete, mactan stone, clay, and PVC, while polyps grown on rubber and steel almost always exhibited the smallest surface-area growth. These differences, however, were not significantly different from each other (Table 1). Comparisons of average performance ranks throughout the 10-wk study period showed concrete-based polyps ranked first (1.87), followed by clay (2.12), mactan stone (3), PVC (3.62), and steel (5). Microcolonies grown on rubber were ranked last (5.375); KW $P = 3.08^5$.

Polyp Production

At the end of the 10-wk study period, microcolonies growing on concrete plates exhibited the highest number of polyps (60.63 polyps, SE = 6.64, $n = 10$), followed by microcolonies growing on clay (60.52 polyps, SE = 8.25, $n = 10$), rubber (58.36 polyps, SE = 10.25, $n = 10$), mactan stone (55.53 polyps, SE = 5.39, $n = 10$), and PVC (55.42 polyps, SE = 7.55, $n = 10$). Microcolonies growing on steel (50.74 polyps, SE = 5.97, $n = 10$) produced the lowest number of polyps (Fig. 4B). These values were not significantly different from each other (ANOVA, $P = 0.93$).

Weekly comparisons of polyp production showed slight fluctuations in the overall performance of microcolonies growing on the six experimental substrates. Concrete- and rubber-grown microcolonies possessed the

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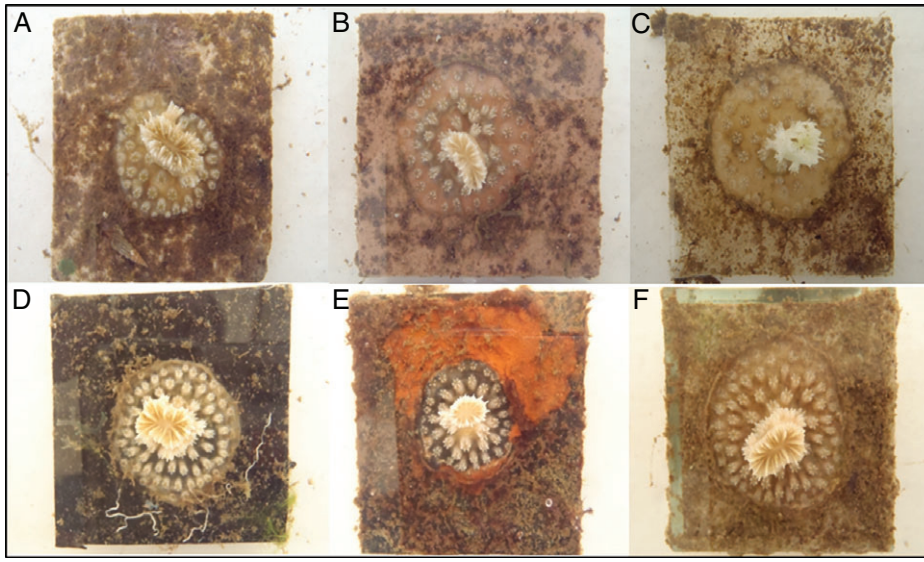

FIGURE 3. *Main polyps and microcolonies of* Galaxea fascicularis *growing on the six experimental substrata after 10 wk. (A) mactan stone, (B) clay, (C) polyvinyl chloride, (D) rubber tire, (E) steel, and (F) concrete.*

largest number of polyps and held a 1–2 position each week, except the ninth week when their positions were reversed. Mactan stone- and clay-grown microcolonies interchanged between third and fourth rankings, while PVC and steel consistently exhibited the smallest number of polyps per week. A comparison of average performance ranks throughout the 10-wk study period shows that concrete-based polyps ranked first (1.25), followed by rubber (2), clay (3.37), mactan stone (3.5), and PVC (5.25). Steel-based polyps were ranked last (5.625); KW $P = 1.47^{-7}$ (Table 2). Comparisons involving weekly polyp production rates, in terms of both weekly addition of polyps and percent increase per week, revealed no significant differences for both within-treatment (same treatment, different weeks) and between-treatment (different treatments, same week) analyses throughout the 10-wk study period.

TABLE 1. *Weekly performance for colony growth of microcolonies grown on the six experimental substrata.*[a]

Week number	Rank 1	2	3	4	5	6	*P* value
1	–	–	–	–	–	–	–
2	–	–	–	–	–	–	–
3	CO	MC	ST	CL	PV	RU	0.9454
4	CL	CO	PV	MC	RU	ST	0.9457
5	CL	MC	CO	ST	PV	RU	0.9563
6	MC	CO	CL	RU	ST	PV	0.9383
7	CO	CL	MC	PV	ST	RU	0.9585
8	CO	CL	PV	MC	RU	ST	0.9192
9	CL	PV	CO	MC	ST	RU	0.8831
10	PV	CO	CL	MC	RU	ST	0.8832

CL = clay; CO = concrete; MC = mactan stone; PV = polyvinyl chloride; RU = rubber; ST = steel.

[a]Basal disk growth was not measurable during the first 2 wk.

Discussion

In the captive husbandry and propagation of stony corals, success is determined by the production of qualitatively and quantitatively superior colonies (Leal et al. 2016). This commonly entails an interplay between the selection of suitable species, morphs, or genetic lines (clones) that are amenable to captive conditions, as well as the provision of physical and chemical conditions that closely mimic or surpass normal environmental conditions that allow the cultured organisms to exhibit increased survival and growth rates. The animal model for the experiment, *G. fascicularis*, is common in the aquarium trade as well as

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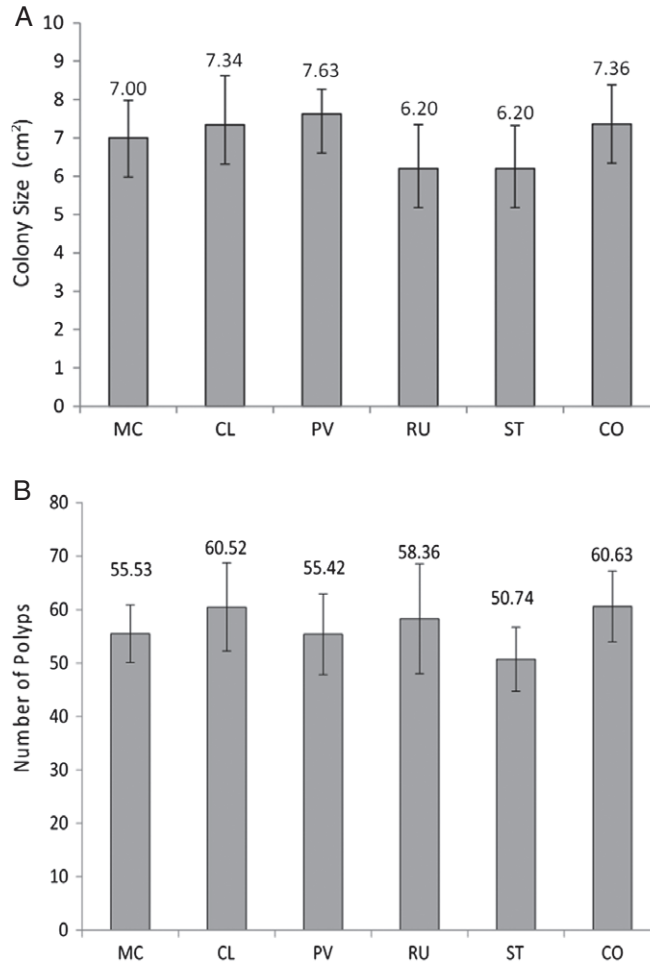

FIGURE 4. *Growth of* Galaxea fascicularis *microcolonies after 10 wk. (A) Colony size (surface area) and (B) number of buds produced. CL = clay; CO = concrete; MC = mactan stone; PV = polyvinyl chloride; RU = rubber tire; ST = steel.*

research, and the results of the experiment demonstrated that different types of substrate material had no differential effect on its survival and growth. The observations were contrary to initial expectations. It was hypothesized that substrate type would differentially affect the survival and growth of *G. fascicularis* microcolonies. This effect, however, was not observed in the results of the experiment. Several factors may have contributed to the observed results. These include (1) the use of *G. fascicularis* as the stony coral species used for the experiment, (2) the biology of fragmented coral colonies, and (3) the nature and design of the experimental setup.

The choice of species for this experiment may have been a factor in the homogeneity of results. *G. fascicularis* was selected as the animal model because of its ease of culture, fast growth, and colony polyp structure that provide uniform replicates for experimentation, making it very suitable for laboratory experiments involving corals (Ferrier-Pages et al. 1998; Schutter et al. 2008, 2011, Hii et al. 2009; Li et al. 2009; Wijgerde et al. 2012a, 2012b). The hardiness of the species, as evidenced by very high survival rates in experiments (Hii et al. 2009; Pillay et al. 2012; van Os et al. 2012), may have allowed it to withstand any chemical leachates or reaction byproducts in the water, while its fast growth

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TABLE 2. *Weekly performance for polyp production of microcolonies grown on the six experimental substrata.*[a]

Week number	Rank 1	2	3	4	5	6	*P* value
1	–	–	–	–	–	–	–
2	–	–	–	–	–	–	–
3	CO	RU	MC	PV	CL	ST	0.505
4	CO	RU	MC	CL	ST	PV	0.565
5	CO	RU	MC	CL	ST	PV	0.733
6	CO	RU	MC	CL	ST	PV	0.699
7	CO	RU	CL	MC	PV	ST	0.621
8	CO	RU	CL	MC	PV	ST	0.845
9	RU	CL	CO	MC	PV	ST	0.835
10	CO	CL	RU	MC	PV	ST	0.937

CL = clay; CO = concrete; MC = mactan stone; PV = polyvinyl chloride; RU = rubber; ST = steel.

[a]Because of tissue inflation, buds could not be distinguished during the first 2 wk.

may have masked any noticeable direct consequential effects of growing on conditions presented by the different experimental substrata. Most notable in this experiment was the ability of the *G. fascicularis* microcolonies to grow easily on steel substrates, something that other coral species like *Acropora palmata* have difficulty doing (Bruckner and Bruckner 2001).

The primary behavior of freshly fragmented coral fragments is to attach, by means of rapid tissue expansion, to any immediate solid surface. Attachment prevents translocation through wave action, leading to abrasion or damage through contact with allelopathic organisms. These usually lead to the death of detached coral fragments (Smith and Hughes 1999; Nagelkerken et al. 2000; Bowden-Kerby 2001; Garrison and Ward 2008). Early reattachment allows fragments to redirect energy to reorganization of polyp orientation, colony shape, growth, and reproduction (Rinkevich 1996; Lindahl 2003). Reattachment requires accelerated rates of tissue expansion compared to normal growth and may override or preclude both chemosensory properties of coral fragments as well as any inhibitory or facilitative effects presented by the growing surface. Similar accelerated rates were observed by Fong and Lirman (1995) in initial tissue regeneration in damaged areas of asexual recruits of *A. palmata* as well as *Acropora cervicornis* (Lirman et al. 2010). This rapid regeneration-directed growth may explain the consistent and uniform rates of tissue expansion of the *G. fascicularis* polyps and microcolonies regardless of substrate type. The isolated polyps may have been aggressively growing in order to provide enough live surface area to attach to the substrate as well as to provide nutrition via heterotrophic feeding and symbiont photosynthesis. Regeneration-driven tissue expansion is sustained by a limited amount of energy (Van Woesik 1998; Lirman 2000) and rapidly slows down with time. This effect seen in the isolated *G. fascicularis* polyps may thus be transitory and may change once the microcolonies reach a certain size, but this was not observed because of the short experimental period.

A note should be made regarding regeneration–growth energy trade-offs as they apply to *G. fascicularis*. Rapid growth of tissue for repair of the damaged area and production of a basal disk for reattachment requires the translocation of energy and negatively affects overall skeletal growth (Meesters et al. 1994; Oren et al. 1997; Lirman 2000). In branching corals, regular linear growth is delayed until regeneration is complete (Lirman and Fong 1997), resulting in an initial slowdown of growth. Effects may be different for nonbranching species and those that employ an encrusting or planar morphology. In these species, peripheral growth onto the substrate is also colony growth, and tissue expansion for regeneration and reattachment follows the same mechanism as normal growth. This may explain the lack of observed slowdown in the growth of the isolated polyps/microcolonies during the course of the experiment.

Coral survival and growth have been documented to be negatively impacted by the presence of heavy metals and petroleum-based products (Guzmán et al. 1991; Haapkylä et al. 2007). Mitchelmore et al. (2007) documented mortalities in *Pocillopora damicornis* exposed to copper and cadmium, while Sabdono (2009) observed similar mortalities in *G. fascicularis*. However, these occurred in experimental conditions in concentrations that may be too high in comparison to levels that would have resulted from the

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natural leaching of the experimental substrata. Shafir et al. (2007) did not observe any mortalities in two species of stony corals, *Stylophora pistillata* and *P. damicornis,* when exposed to water-soluble fractions of crude oil and seawater, indicating that very low and weak concentrations may have very little effect on coral colonies. Although substrate–seawater reactions were observed (rusting of steel, oily sheen in rubber tire treatments, reddish tint in clay treatments), the flow-through, high aeration design of the system may have prevented the buildup of these leachates to levels high enough to cause observable behavioral or physiological tissue damage or mortalities. The high water turnover and efficient aeration of the utilized experimental system may have also prevented reaction-based swings or changes in the pH and dissolved oxygen levels on the system water that could have affected the survival and growth rates.

In the context of coral culture, the most important result of this work is that survival and growth of fragments of the test coral were not affected by substratum type and occurred at consistent, equal rates despite the presumed positive and negative properties of the experimental substrata. This finding may have important implications for the culture of coral fragments by addressing the question of the best material to use as a culture substrate. No single material has properties that produce qualitatively and quantitatively superior colonies, and corals will grow at the same rates despite differences in substratum material. Thus, practitioners can theoretically choose whatever material satisfies their limitations and requirements. With regard to the aspect of coral reef remediation, it raises the possibility that nonperforming artificial reef structures that fail to attract chemosensitive larvae can be rehabilitated using less-sensitive coral fragments as transplant material that will grow regardless of the makeup of the artificial reef structure.

These experimental results have limitations, and the absence of predicted dissimilar effects of the selected substrates needs to be further tested. For example, the use of a single species for this experiment fails to establish a universal effect on all coral species. Coral species vary in their responses to various environmental factors (Yap et al. 1992; Hall 1997; Smith and Hughes 1999; Yap 2004), and the use of different species and growth forms could reveal growth patterns and behaviors not observed in this work. The duration of the experiment requires further evaluation. The 10-wk period may not have been long enough to allow the observation of latent or delayed reactions between the substrata and seawater, as well as effects of long-term exposure/contact between the reaction byproducts on the test organism. The possible effects of aquaculture system and culture method also need to be addressed. Potential effects of leachates and reaction products may be affected by varying flow, circulation levels, and culture systems. Critical buildup of potential toxins may not occur in high-flow and flow-through systems or in open-ocean mariculture but may occur and be affective in low-flow and recirculating setups, such as those employed in inland aquarium culture systems.

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